

VAANI GOENKA

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EDUCATION

Ashoka University | 2024-2028

Major: Computer Science, Minor: Mathematics | **Received a 100% merit scholarship for Computer Science (given to 20-25 students across the batch)**

CGPA: 3.95 / 4.00

Relevant coursework: Calculus, Multivariate Calculus, Discrete Mathematics, Probability & Statistics, Introduction to Computer Science, Cryptography, Data Structures & Algorithms, Programming Lab, Symbolic Logic and Applications

Modern High School for Girls | 2010 - 2024

ICSE: 97% (Mar 2022) | ISC: 98.25% (Mar 2024), School Rank 4 | **General Proficiency award** (Highest Academic Award – Sole recipient)

SAT Examination: 1580/1600

Advanced Placement (AP) Examinations: Computer Science A: 5, Calculus BC: 4, Calculus AB (subscore): 5, English Language and Composition: 5

EXPERIENCE

Research Assistant, *MPhasis Lab, "AI At Ashoka"*

May 2025 - Current

- Co-authored "BOOP: Write Right Code", a research paper on CS Pedagogy and Program Correctness, accepted at ACM COMPUTE 2025.
- Developed a Visual Studio Extension, leveraging programming language parsers, automated theorem provers and on-ground data. Moreover, developed instructor and student interfaces for optimal adoption.
- Currently working on automated grading using automated tools for verifying code and related formal methods.

Teaching Assistant, *Ashoka University*

May 2025 - Dec 2025

- Led weekly tutorials and review sessions for students across multiple advanced CS and Mathematics courses.
- Previously a teaching Assistant in the Lodha Genius Program for "Introduction to Modern Cryptography", "Linear Algebra", "Graph Theory & Recurrence Relations" and "Problem Solving Group" (Mathematics Olympiad Style Questions).
- Previously a teaching Assistant to Prof. Gaurav Bhatnagar for "Discrete Mathematics" (undergraduate course).

Yugen Space, *Software Engineering Intern - LLMs and AI*

December 2024 - February 2025

- Developed pipelines for RAG systems, database integration, and frontend workflows for an AI-integrated Geospatial Data Platform.
- Built geo-spatial data visualization with LLM-enabled interaction, enabling multimodal (limited) queries.

Ashoka University Ministry of Technology, *Deputy Minister*

August 2024 - Ongoing

- Designed, developed and deployed an end-to-end solution for automating the Mailroom package system at Ashoka University – currently in use by 4000 students, 10 staff members, helping them reduce time spent per package by 70%; **received a 'Student Life Service' Award**. Developed a gamified notification system; responsible for ongoing server management and maintenance.
- Led a team of 10 developers to build micro and macro digital infrastructure for the student body and maintain the systems in place, working closely with the IT and Administration teams.
- Revamped legacy platforms using a full-stack React/Next.js/Strapi framework, improving design and UI while incorporating iterative user feedback.

AWARDS & HONORS

Runner-Up | Apart Research Sprint (Artificial Intelligence and AI Safety)

2025

Excellence in Computer Science, Psychology, Hindi, English, Physics

2024

AP Scholar

2023

Excellence in Information & Communication Technology

22-23, 23-24

Inter-School Coding Competitions: Podium finisher at 7 inter-school coding competitions

22-23

All Rounder Achievement: for excellence in both academic and non-academic fields

22-23

1st Place | National Space Science Symposium

2021

INITIATIVE & LEADERSHIP

Mentica (Mental health website and movement), *Founder and Developer* | [Covered by the Times of India](#)

May 2023 - May 2024

- Developed a quiz-system probability assessment for potential mental-health difficulties faced by users (in accordance with professional opinion and DSM guidelines) and collated 150+ resources for conditions-even specific difficulties, a unique online-journaling system and provisions for professional outreach. Reached 15,000+ visits, 12,000+ quizzes taken, 5000+ repeat resource users, 7000+ views on the blog.

Student Council, *Head of Technology*

April 2023 - 2024

- The only student with 6 positions of leadership. Developed the first-ever secure and fully responsive website for our school's inter-school fest which digitized the tedious process of pen and paper registration of schools, increasing reach by **30,000%** in a month.
- Founded the first coding club in the school (also the first student-found club). Increased participation in coding competitions by 1270%, mentored 50+ students.

Accelerated Mathematics, *Student*

April 2023 - 2024

- Selected for advanced mathematics training in school. Engaged in olympiad-style problem solving and real-world applications of mathematics.